Analisi e Progetto di Algoritmi

Laurea Triennale in Informatica
(codice: E3101Q113)

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Longest Common Subsequence (LCS) con al massimo 3 elementi rossi

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Problema LCS(X,Y,3)

INPUT:

- $X = \langle x_1, x_2, \ldots, x_m \rangle$
- $Y = \langle y_1, y_2, \ldots, y_n \rangle$
- **col** : $\Sigma \rightarrow$ {"red", "black"}
- **3**, numero di elementi rossi che al massimo devono essere presenti

OUTPUT:

- **LCS(X,Y,3)**, LCS con al massimo **3** elementi rossi

Problema LCS(X,Y,3)

$X = \langle 3, 5, 9, 6, 4, 8, 12, 10, 5, 30, 7 \rangle$

$Y = \langle 3, 2, 4, 9, 6, 10, 2, 8, 30, 13, 30, 7 \rangle$

$LCS(X,Y) = \langle 3, 9, 6, 8, 30, 7 \rangle$ **or** $\langle 3, 9, 6, 10, 30, 7 \rangle$

Problema LCS(X,Y,3)

X = ⟨3, 5, 9, 6, 4, 8, 12, 10, 5, 30, 7⟩

Y = ⟨3, 2, 4, 9, 6, 10, 2, 8, 30, 13, 30, 7⟩

LCS(X,Y) = ⟨3, 9, 6, 8, 30, 7⟩ or ⟨3, 9, 6, 10, 30, 7⟩

Problema LCS(X,Y,3)

$X = \langle 3, 5, 9, 6, 4, 8, 12, 10, 5, 30, 7 \rangle$

$Y = \langle 3, 2, 4, 9, 6, 10, 2, 8, 30, 13, 30, 7 \rangle$

$LCS(X,Y) = \langle 3, 9, 6, 8, 30, 7 \rangle$ or $\langle 3, 9, 6, 10, 30, 7 \rangle$

$LCS(X,Y,3) = \langle 3, 6, 8, 30, 7 \rangle$

Soluzione con DP

1. Formulazione ricorsiva
   I. Individuazione dei sottoproblemi e dei coefficienti
   II. Individuazione del valore ottimo
   III. Determinazione delle equazioni di ricorrenza
      a. Casi base
      b. Passo ricorsivo
2. Algoritmo bottom-up per il calcolo del valore ottimo
3. Algoritmo ricorsivo di ricostruzione di LCS(X,Y,3)

Sottoproblema

Sottoproblema di dimensione (i, j, r)

Trovare la LCS dei prefissi X_i e Y_j con al massimo r elementi rossi → **$LCS(X_i, Y_j, r)$**

$i \in \{0, 1, \dots, m\}$

$j \in \{0, 1, \dots, n\}$

$r \in \{0, 1, \dots, 3\}$

Numero totale di sottoproblemi: $(m+1)(n+1)(3+1)$

Problema principale ≡ Sottoproblema di dimensione (m,n,3)

Coefficienti

Coefficiente $c_{i,j,r}$ del sottoproblema di dimensione (i, j, r)

$c_{i,j,r}$ è la lunghezza della LCS dei prefissi X_i e Y_j con al massimo r elementi rossi

$i \in \{0, 1, \ldots, m\}$

$j \in \{0, 1, \ldots, n\}$

$r \in \{0, 1, \ldots, 3\}$

Numero totale di coefficienti: $(m+1)(n+1)(3+1)$

Valore ottimo del problema principale $\equiv c_{m,n,3}$

Eq. ricorrenza - Casi base

$(i \geq 0 \land j = 0) \land r = 0$

$\Rightarrow LCS(X_i, Y_j, r) = \langle\rangle$

$\Rightarrow \mathbf{c_{i,j,r} = 0}$

$(i = 0 \land j > 0) \land r = 0$

$\Rightarrow LCS(X_i, Y_j, r) = \langle\rangle$

$\Rightarrow \mathbf{c_{i,j,r} = 0}$

Eq. ricorrenza - Casi base

$(i \geq 0 \wedge j = 0) \wedge r > 0$

$\Rightarrow LCS(X_i, Y_j, r) = \langle\rangle$

$\Rightarrow c_{i,j,r} = 0$

$(i = 0 \wedge j > 0) \wedge r > 0$

$\Rightarrow LCS(X_i, Y_j, r) = \langle\rangle$

$\Rightarrow c_{i,j,r} = 0$

Casi base (in sintesi)

$(i = 0 \lor j = 0) \land r \geq 0$

$\Rightarrow LCS(X_i, Y_j, r) = \langle\rangle$

$\Rightarrow$ **$c_{i,j,r} = 0$**

Eq. ricorrenza – Passo ricorsivo

$X_i = \langle x_1, x_2, \ldots, x_{i-1}, x_i \rangle$

$Y_j = \langle y_1, y_2, \ldots, y_{j-1}, y_j \rangle$

$(i > 0 \wedge j > 0)$

Eq. ricorrenza – Passo ricorsivo

$X_i = \langle x_1, x_2, \ldots, x_{i-1}, x_i \rangle$

$Y_j = \langle y_1, y_2, \ldots, y_{j-1}, y_j \rangle$

$(i > 0 \land j > 0) \land r = 0$

Eq. ricorrenza – Passo ricorsivo

$X_i = \langle x_1, x_2, \ldots, x_{i-1}, x_i \rangle$

$Y_j = \langle y_1, y_2, \ldots, y_{j-1}, y_j \rangle$

$(i > 0 \land j > 0) \land r = 0$

$x_i \neq y_j$

$\Rightarrow LCS(X_i, Y_j, r) = \max\{LCS(X_{i-1}, Y_j, r), LCS(X_i, Y_{j-1}, r)\}$

$\Rightarrow \mathbf{c_{i,j,r} = \max\{c_{i-1,j,r}, c_{i,j-1,r}\}}$

Eq. ricorrenza – Passo ricorsivo

$X_i = \langle x_1, x_2, \ldots, x_{i-1}, x_i \rangle$

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$\Rightarrow \mathbf{c_{i,j,r} = \max\{c_{i-1,j,r}, c_{i,j-1,r}\}}$

$x_i = y_j$

Eq. ricorrenza – Passo ricorsivo

$X_i = \langle x_1, x_2, \ldots, x_{i-1}, x_i \rangle$
$Y_j = \langle y_1, y_2, \ldots, y_{j-1}, y_j \rangle$

$(i > 0 \wedge j > 0) \wedge r = 0$

- **$x_i \neq y_j$**
  - $\Rightarrow LCS(X_i, Y_j, r) = \max\{LCS(X_{i-1}, Y_j, r), LCS(X_i, Y_{j-1}, r)\}$
  - $\Rightarrow$ **$c_{i,j,r} = \max\{c_{i-1,j,r}, c_{i,j-1,r}\}$**
- **$x_i = y_j$**
  - **$col(x_i) \neq$ "red"**
    - $\Rightarrow LCS(X_i, Y_j, r) = LCS(X_{i-1}, Y_{j-1}, r) + \langle x_i \rangle$
    - $\Rightarrow$ **$c_{i,j,r} = c_{i-1,j-1,r} + 1$**

Eq. ricorrenza – Passo ricorsivo

$X_i = \langle x_1, x_2, \ldots, x_{i-1}, x_i \rangle$
$Y_j = \langle y_1, y_2, \ldots, y_{j-1}, y_j \rangle$

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$X_i = \langle x_1, x_2, \ldots, x_{i-1}, x_i \rangle$

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- **$x_i = y_j$**
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$col(x_i) =$ "red"

$\Rightarrow LCS(X_i, Y_j, r) = LCS(X_{i-1}, Y_{j-1}, r-1) + \langle x_i \rangle$

$\Rightarrow \mathbf{c_{i,j,r} = c_{i-1,j-1,r-1} + 1}$

Passo ricorsivo (in sintesi)

$i > 0 \wedge j > 0$

$x_i \neq y_j$

$\Rightarrow LCS(X_i, Y_j, r) = \max\{LCS(X_{i-1}, Y_j, r), LCS(X_i, Y_{j-1}, r)\}$

$\Rightarrow \mathbf{c_{i,j,r} = \max\{c_{i-1,j,r}, c_{i,j-1,r}\}}$

$x_i = y_j \wedge col(x_i) \neq$ "red"

$\Rightarrow LCS(X_i, Y_j, r) = LCS(X_{i-1}, Y_{j-1}, r) + \langle x_i \rangle$

$\Rightarrow \mathbf{c_{i,j,r} = c_{i-1,j-1,r} + 1}$

$x_i = y_j \wedge col(x_i) =$ "red" $\wedge\ r = 0$

$\Rightarrow LCS(X_i, Y_j, r) = LCS(X_{i-1}, Y_{j-1}, r)$

$\Rightarrow \mathbf{c_{i,j,r} = c_{i-1,j-1,r}}$

$x_i = y_j \wedge col(x_i) =$ "red" $\wedge\ r > 0$

$\Rightarrow LCS(X_i, Y_j, r) = LCS(X_{i-1}, Y_{j-1}, r-1) + \langle x_i \rangle$

$\Rightarrow \mathbf{c_{i,j,r} = c_{i-1,j-1,r-1} + 1}$

Soluzione del problema (DP)

1. Algoritmo bottom-up per il calcolo di $c_{m,n,3}$ (valore ottimo)
    - memorizzazione dei coefficienti

2. Algoritmo ricorsivo per ricostruire LCS(X,Y,3)

Algoritmo bottom-up (DP)

- Costruzione di una matrice C di dimensione $(m+1)(n+1)(3+1)$

- $C[i, j, r] \equiv c_{i,j,r}$

- $C[m, n, 3]$ è il valore ottimo del problema

Algoritmo bottom-up (DP)

```
int calcola_ottimo_LCS_max_3(X,Y)
    C ← matrice di dimensione (m+1)(n+1)(3+1)
    //Casi base per (i ≥ 0 ∧ j = 0) Λ r ≥ 0
    for r from 0 to 3 do
        for i from 0 to m do
            C[i,0,r] ← 0
    //Casi base per (i = 0 ∧ j > 0) Λ r ≥ 0
    for r from 0 to 3 do
        for j from 1 to n do
            C[0,j,r] ← 0
```

Algoritmo bottom-up (DP)

```
[…]
//Passo ricorsivo per (i > 0 ∧ j > 0) ∧ r ≥ 0
for r from 0 to 3 do
  for i from 1 to m do
    for j from 1 to n do
      if xi ≠ yj then
        C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}
      else        //xi = yj
        if col(xi) ≠ "red" then
          C[i,j,r] = C[i-1,j-1,r] + 1
        else    //col(xi) = "red"
          if r = 0 then
            C[i,j,r] = C[i-1,j-1,r]
          else        //r > 0
            C[i,j,r] = C[i-1,j-1,r-1] + 1
return C[m,n,3]
```

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
```

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
    if i > 0 and j > 0 then
```

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
   if i > 0 and j > 0 then
     if x_i ≠ y_j then
        C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}
     else     //x_i = y_j
       if col(x_i) ≠ "red" then
         C[i,j,r] = C[i-1,j-1,r] + 1
       else   //col(x_i) = "red"
         if r = 0 then
           C[i,j,r] = C[i-1,j-1,r]
         else       //r > 0
           C[i,j,r] = C[i-1,j-1,r-1] + 1
```

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
    if i > 0 and j > 0 then
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        if col(x_i) ≠ "red" then
           C[i,j,r] = C[i-1,j-1,r] + 1
        else   //col(x_i) = "red"
          if r = 0 then
             C[i,j,r] = C[i-1,j-1,r]
          else       //r > 0
             C[i,j,r] = C[i-1,j-1,r-1] + 1
```

$\Rightarrow$ LCS(X_i,Y_j,r) = LCS(X_{i-1},Y_{j-1},r) + $\langle x_i \rangle$

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
    if i > 0 and j > 0 then
      if xi ≠ yj then
          C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}
      else    //xi = yj
        if col(xi) ≠ "red" then
          Print_LCS_3(C, X, Y, i-1, j-1, r); print xi
        else   //col(xi) = "red"
          if r = 0 then
            C[i,j,r] = C[i-1,j-1,r]
          else      //r > 0
            C[i,j,r] = C[i-1,j-1,r-1] + 1
```

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
    if i > 0 and j > 0 then
      if xi ≠ yj then
          C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}
      else     //xi = yj
        if col(xi) ≠ "red" then
          Print_LCS_3(C, X, Y, i-1, j-1, r); print xi
        else   //col(xi) = "red"
          if r = 0 then
            C[i,j,r] = C[i-1,j-1,r]
          else       //r > 0
            C[i,j,r] = C[i-1,j-1,r-1] + 1
```

$\Rightarrow$ LCS(X_i,Y_j,r) = LCS(X_{i-1},Y_{j-1},r-1) + $\langle x_i \rangle$

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
    if i > 0 and j > 0 then
      if xi ≠ yj then
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        if col(xi) ≠ "red" then
           Print_LCS_3(C, X, Y, i-1, j-1, r); print xi
        else   //col(xi) = "red"
          if r = 0 then
             C[i,j,r] = C[i-1,j-1,r]
          else      //r > 0
             Print_LCS_3(C, X, Y, i-1, j-1, r-1); print xi
```

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
    if i > 0 and j > 0 then
      if xi ≠ yj then
          C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}
      else     //xi = yj
        if col(xi) ≠ "red" then
          Print_LCS_3(C, X, Y, i-1, j-1, r); print xi
        else    //col(xi) = "red"
          if r = 0 then
            C[i,j,r] = C[i-1,j-1,r]  ⇒ LCS(Xi,Yj,r) = LCS(Xi-1,Yj-1,r)
          else       //r > 0
            Print_LCS_3(C, X, Y, i-1, j-1, r-1); print xi
```

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
    if i > 0 and j > 0 then
      if xi ≠ yj then
           C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}
      else      //xi = yj
        if col(xi) ≠ "red" then
           Print_LCS_3(C, X, Y, i-1, j-1, r); print xi
        else   //col(xi) = "red"
           if r = 0 then
             Print_LCS_3(C, X, Y, i-1, j-1, r)
           else        //r > 0
             Print_LCS_3(C, X, Y, i-1, j-1, r-1); print xi
```

Ricostruzione

if C[i,j,r] = C[i-1,j,r] **then**

  $LCS(X_i, Y_j, r) = LCS(X_{i-1}, Y_j, r)$

else   // C[i,j,r] = C[i,j-1,r]

  $LCS(X_i, Y_j, r) = LCS(X_i, Y_{j-1}, r)$

procedura Print_LCS_max_3(C,

  if i > 0 and j > 0 **then**

   if $x_i \neq y_j$ **then**

    ~~C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}~~

   else   //$x_i = y_j$

    if col(x_i) ≠ “red” **then**

     Print_LCS_3(C, X, Y, i-1, j-1, r); print x_i

    else //col(x_i) = “red”

     if r = 0 **then**

      Print_LCS_3(C, X, Y, i-1, j-1, r)

     else   //r > 0

      Print_LCS_3(C, X, Y, i-1, j-1, r-1); print x_i

Ricostruzione

if C[i,j,r] = C[i-1,j,r] **then**
  Print_LCS_3(C, X, Y, i-1, j, r)
else   // C[i,j,r] = C[i,j-1,r]
  LCS(X_i,Y_j,r) = LCS(X_i,Y_{j-1},r)

procedura Print_LCS_max_3(C,
  if i > 0 and j > 0 **then**
    if $x_i \neq y_j$ **then**
      ~~C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}~~
    else   //$x_i = y_j$
      if col(x_i) ≠ "red" **then**
        Print_LCS_3(C, X, Y, i-1, j-1, r); print x_i
      else   //col(x_i) = "red"
        if r = 0 **then**
          Print_LCS_3(C, X, Y, i-1, j-1, r)
        else   //r > 0
          Print_LCS_3(C, X, Y, i-1, j-1, r-1); print x_i

Ricostruzione

```
if C[i,j,r] = C[i-1,j,r] then
  Print_LCS_3(C, X, Y, i-1, j, r)
else      // C[i,j,r] = C[i,j-1,r]
  Print_LCS_3(C, X, Y, i, j-1, r)
```

```
procedura Print_LCS_max_3(C,
    if i > 0 and j > 0 then
      if x_i ≠ y_j then
          C[i,j,r] = max{C[i-1,j,r], C[i,j-1,r]}
      else     //x_i = y_j
        if col(x_i) ≠ "red" then
          Print_LCS_3(C, X, Y, i-1, j-1, r); print x_i
        else    //col(x_i) = "red"
          if r = 0 then
            Print_LCS_3(C, X, Y, i-1, j-1, r)
          else        //r > 0
            Print_LCS_3(C, X, Y, i-1, j-1, r-1); print x_i
```

Ricostruzione di LCS(X_i,Y_j,r)

```
procedura Print_LCS_max_3(C, X, Y, i, j, r)
    if i > 0 and j > 0 then
      if xi ≠ yj then
       if C[i,j,r] = C[i-1,j,r] then
          Print_LCS_3(C, X, Y, i-1, j, r)
       else    // C[i,j,r] = C[i,j-1,r]
          Print_LCS_3(C, X, Y, i, j-1, r)
      else    //xi = yj
        if col(xi) ≠ "red" then
          Print_LCS_3(C, X, Y, i-1, j-1, r); print xi
        else   //col(xi) = "red"
          if r = 0 then
            Print_LCS_3(C, X, Y, i-1, j-1, r)
          else       //r > 0
            Print_LCS_3(C, X, Y, i-1, j-1, r-1); print xi
```